

MID-TERM EXAM

Macro M1 - 2015

Ex 1 (1) 4 pts (+0.5 Bonus)

14 pts + 0.5 + 0.5
→ 15 pts

1 pt { A competitive equilibrium is a system of prices $\{P_1, P_2, w\}$ and real allocations $\{c_1, c_2, n_2, g_2\}$ such that:

1 pt { 1) The consumer maximizes the intertemporal (the representative household) utility given prices and taxes, subject to the intertemporal budget constraint

1 pt { 2) The firms maximizes profit, given prices and technology

(0.5 pt Bonus) 3) The government satisfies the intertemporal budget constraint

1 pt { 4) Markets clear (supply = Demand for each market)
$$c_1 = \bar{y}$$
$$c_2 + g_2 = y_2 (\equiv n_2)$$
$$\text{Labor Demand} = \text{Labor supply}$$

② 4 pto (+ 0.5 Bonus)

Household problem

$$1^{ph} \left\{ \begin{array}{l} \frac{1}{c_1} = \beta \frac{P_1}{P_2} \frac{1}{c_2} \\ \beta = \frac{W}{P_2} \frac{\beta}{c_2} \end{array} \right.$$

Firm's problem

Firm's problem

1st } FOC $P_2 = w \quad (\Leftrightarrow \frac{w}{P_2} = 1)$

$\hookrightarrow \text{profit} = 0 \quad (\text{CRS})$

MRS consumption / leisure

$$\beta = \frac{\beta}{c_2}$$

$$1st \quad \{ \Rightarrow c_2 = 1$$

1pr { Period 1 $C_1 = \bar{y}$
Period 2 $1 + g_2 = y_2 = n_2$

0.5 pt (Bonus) $\left\{ \frac{P_2}{P_1} = \beta \frac{C_1}{C_2} = \bar{y} \right.$

3/ 2 pts

Central Planner (no prices)

$$\left\{ \begin{array}{l} \max \log c_1 + \beta \log c_2 - \beta n_2 \\ \text{s.t. } c_1 = \bar{y} \\ c_2 + g_2 = y_2 = n_2 \\ \hookrightarrow c_2 = n_2 - g_2 \end{array} \right.$$

$$\left. \begin{array}{l} \text{1pt} \\ \text{FOC} \end{array} \right\} \begin{array}{l} \frac{\cancel{\beta}}{n_2 - g_2} = \beta \\ n_2 - g_2 = 1 \\ n_2 = 1 + g_2 = y_2 \\ \underline{\text{so}} \quad c_2 = 1 \\ c_1 = \bar{y} \end{array}$$

the same real allocations.

4) 2 pts

$$\left. \begin{array}{l} \text{1pt} \end{array} \right\} \begin{array}{l} g_2 \uparrow \rightarrow \text{no effect on } c_2 \\ \rightarrow n_2 \uparrow \\ \rightarrow y_2 \uparrow \end{array}$$

1pt } People work more because they anticipate a negative wealth effect. The multiplier is equal to unity. No crowding-out effect

5 / 2pts

Ricardian equivalence

1pt { t_1, t_2 no effect on real allocations
and prices

In other words

1pt { t_1^*, t_2^* with $t_1^* \neq t_1$ and $t_2^* \neq t_2$
and t_1^*, t_2^* satisfy the intertemporal
budget constraint of the government produce
the same real allocations and prices
 $\Rightarrow$ the timing of taxes does not matter

Exercise 2

① 3 pts

$$1 \text{ pt} \left\{ \begin{array}{l} \max E_t \sum_{i=0}^{\infty} \beta^i U(c_{t+i}) \\ \text{s.t. } A_{t+1} = (1+r)A_t + y_{t+1}^d - c_t \quad \forall t \\ \text{with } U(c_t) = \alpha_0 c_t - \frac{\alpha_1}{2} c_t^2 \end{array} \right.$$

$$2 \text{ pts} \left\{ \begin{array}{l} \text{FOC } U'(c_t) = \beta(1+r) E_t U'(c_{t+1}) \\ \text{with } U'(c_t) = \alpha_0 - \alpha_1 c_t \\ \beta(1+r) = 1 \\ \hookrightarrow E_t c_{t+1} = c_t \end{array} \right.$$

② 3 pts

$$r > 0 \Rightarrow 1+r > 1$$

$\hookrightarrow$ Forward substitution

$$1 \text{ pt} \left\{ \begin{array}{l} A_t = \frac{1}{1+r} E_t \sum_{i=0}^{\infty} \left(\frac{1}{1+r} \right)^i c_{t+i} - \frac{1}{1+r} E_t \sum_{i=0}^{\infty} \left(\frac{1}{1+r} \right)^i y_{t+i}^d \\ \text{where we have imposed the transversality condition.} \\ \lim_{T \rightarrow \infty} \left(\frac{1}{1+r} \right)^T E_t A_{t+T} = 0 \end{array} \right.$$

4 / 2 pts

illustration of the Lucas critique with
two polar cases.

$$\begin{array}{l} 1pt \left\{ \begin{array}{l} \boxed{\ell=0} \quad \frac{\partial C_t}{\partial \tau} = -\frac{r}{1+r} \bar{Y} \quad \text{given } \bar{Y}, \text{ small effect} \\ \boxed{\ell=1} \quad \frac{\partial C_t}{\partial \tau} = -\bar{Y} \quad \text{large} \end{array} \right. \end{array}$$

1pt $\left\{ \begin{array}{l} \text{Permanent income argument} \quad \text{Consumption} \\ \text{is more sensitive to permanent change} \\ \text{in tax rate than transitory changes} \end{array} \right.$